

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 39

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Warming melts ice shelves that hold back glaciers and lubricates ice flow so land ice reaches the sea faster (Antarctic Peninsula most affected); sea ice loss removes krill habitat and lowers albedo • Raised UV harms planktonic algae, krill and fish / sea urchin larvae in surface waters (land animals protected by fur / feathers) • Disturbance of wildlife, introduction of non-indigenous species and pathogens, fuel and sewage pollution; visitor numbers rising rapidly at a few landing sites • Reserves of coal, oil, gold, uranium, platinum etc; pressure to exploit will rise as reserves elsewhere deplete though currently banned and operations extremely difficult	4
01.2	• Treaty signed by 52 countries including all active there; no military activity (except supporting research), no nuclear explosions or radioactive waste dumping • No commercial mineral exploitation • Fisheries regulated by CCAMLR: krill and some fish exploited, fisheries closed if over-exploited; no commercial exploitation of mammals • Tourism control: no accommodation on land (wastes stay on ships), one ship per site with maximum 100 ashore, trained guides, keep 5 m from wildlife, no litter, cleaned protective clothing to prevent introductions • Waste management: all wastes removed (fuel containers, batteries, plastics, oil); sewage only discharged where sea currents disperse it; old waste removed unless removal causes more damage • Only long-term residents are research scientists and support staff; strict regulation of research impacts	6
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 69-71; AQA spec 3.1.2.3.4.	
02.1	• Yield increases as inputs increase • But each extra unit of input produces a smaller increase in yield than the previous one (the curve flattens / law of diminishing returns)	2

Question	Marking guidance	Marks
02.2	• Second unit: $70 - 50 = 20$ units • Fifth unit: $94 - 90 = 4$ units	2
02.3	• The first units of input on each area give the largest yield increases • Concentrating inputs means later units add very little, whereas the same inputs spread thinly all work at the steep part of the curve	2
02.4	• Some areas have much greater potential productivity so investment there is sensible • Affluent societies can afford to buy more inputs while poorer communities that would benefit most cannot afford them (economic reasons)	2
02.5	• Yield per unit area is very high because of large inputs • But the extra yield per unit of input is small, so efficiency (yield per unit input / energy ratio) is low	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 324-325; AQA spec 3.5.1.2.	
03.1	• Trays of the same soil (texture, compaction, organic matter, depth) with different percentage vegetation cover (eg 0, 25, 50, 75, 100% grass), replicated • Same slope / gradient; simulated rain from a standard height at a standard rate and drop size (eg watering can / sprinkler) for a standard time • Collect the soil splashed / washed off (eg onto a sheet or into a trough at the base), dry and weigh it • Repeat each treatment several times and calculate means (and standard deviation) • Plot soil loss against cover; statistical test (Spearman's rank / t-test) for significance	5
03.2	<i>Any 3 from:</i> • Rainfall intensity / raindrop size / height • Soil texture, compaction, organic matter, depth • Permeability of the material beneath • Gradient • Root binding	3
03.3	• All variables except the independent variable can be standardised in the laboratory; in the field uncontrollable variables (weather) must be measured instead • Laboratory conditions may not represent natural conditions so results may not apply directly to real ecosystems	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 419-420; AQA spec 3.7.1, 3.7.2.2.	
04.1	• General physiological solvent: most biological reactions take place between substances dissolved in water • Transport medium in blood / sap carrying gases, sugars, amino acids, nutrients, wastes • High specific heat capacity moderates temperature changes; evaporation absorbs heat (cooling) • Anomalous expansion: ice floats so water bodies do not freeze solid; absorbs UV protecting early marine life; provides aquatic habitats	4

Question	Marking guidance	Marks
04.2	• From about 2.7 billion years ago photosynthetic archaea / bacteria (then algae and plants) released oxygen; at first it reacted with dissolved iron (forming iron ore deposits) then built up in the oceans and atmosphere (from about 2.45 billion years ago) • Oxygen in the stratosphere absorbed UV, forming ozone (O₃) in a dynamic equilibrium of O, O₂ and O₃ • The ozone layer reduced UV reaching the surface so organisms could survive out of the water • Oxygen allowed aerobic respiration releasing more energy so complex multicellular organisms evolved; many anaerobic archaea died out	4
04.3	• Photosynthesis removed CO₂ (a greenhouse gas) from the atmosphere, storing carbon in sediments, carbonate rocks and fossil fuels • This reduced infrared retention, offsetting the increasing brightness of the Sun (about 10% per billion years) so temperatures did not rise excessively	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 7-8; AQA spec 3.1.1.1-3.1.1.2.	
05.1	• A community maintained at a stage before the climax by regular human activity (grazing, mowing, burning, coppicing) that deflects succession • Chalk grassland was produced by centuries of sheep grazing; without it succession to scrub and woodland resumes	2
05.2	• Grazing removes taller / dominant grasses and prevents scrub establishing (5% cover) so small plants are not shaded out • Trampling creates bare ground for wildflower seed germination and warm patches for invertebrates / reptiles • Dung creates small nutrient-rich patches supporting different plants and invertebrates; sheep eat short grasses and some bushes • Mowing removes everything at once including plants grazers avoid, and removes flowers / seeds before setting; burning removes vegetation but also kills invertebrates and allows scrub regrowth; abandoned plot dominated by scrub (55%) so few species	4
05.3	• Sheep eat short grass (some breeds eat scrub); ponies avoid flowers but eat bracken; cattle remove long rough grass • Choice depends on the vegetation to be controlled and the plants to be protected	2
05.4	• Removes mature heather bushes and young trees that grazers will not eat, creating open unshaded ground • Heat stimulates dormant heather seeds to germinate so young heather re-establishes	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 47-49; AQA spec 3.1.3.3.	
06.1	• South America: high loss (5.1-5.5 Mha yr⁻¹) falling by about half by 2020 • Africa: loss steady at about 3.4 then rising to 3.9 by 2020 - now the highest • Europe: negative values - forest area increasing (afforestation exceeds loss), though the gain is shrinking	3

Question	Marking guidance	Marks
06.2	• South America: clearance for commercial cash crops (sugar cane, soya) and cattle ranching for export; HEP reservoirs; roads improving access; mining • Africa: growing rural populations clearing land for subsistence farming; fuelwood and charcoal collection above regrowth (charcoal wastes much of the wood's energy); fodder cutting in dry areas • Both: hardwood timber (teak, mahogany) from clear-felled mixed forest often not replanted; urban expansion	4
06.3	• They create barriers fragmenting wildlife populations • Improved access makes it easier to remove timber and farm produce, so clearance spreads along them	2
06.4	• Conifers grow quickly so plantations are replanted after harvest and yield a saleable crop within decades; slow-growing hardwoods (teak, mahogany) take too long for commercial plantations so they are taken from natural forest that is clear-felled and not replanted	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 367-368; AQA spec 3.5.3.3.	
07.1	<i>Any 2 from:</i> • Hedgerows • Hay meadows • Grazed moorland / heathland • Chalk grassland	2
07.2	• Drainage produces more aerobic soils so wetland plants and the animals they support cannot survive • Fertilisers increase growth of tall species such as grasses which out-compete smaller plants such as wildflowers	2
07.3	• The introduced (non-indigenous) predator may eat other species as well as the pest	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-329; AQA spec 3.5.1.3.	
08.1	• Its reading is adjusted to match a meter known to be accurate so that the results are accurate / comparable • If it cannot be adjusted the percentage error should be recorded so the results can be corrected	2
08.2	<i>Any 3 from:</i> • Droppings: diet, gender, territory size • Owl pellets: diet • Feeding marks (eg chewed hazel nuts): presence of species such as dormice • Nests / burrows: presence and breeding • Tracks / footprints: presence, movements • Territorial marks (scratching posts): presence and territory	3
08.3	• Bat detectors / sonograms record the ultrasonic sounds bats produce, which differ between species • The recordings provide evidence of presence, abundance and activity of species that cannot easily be seen	2

Question	Marking guidance	Marks
08.4	<i>Any 3 from:</i> • Marked individuals mix freely with the rest of the population • Marks do not fall off or wear away, and marking does not affect survival • The population does not change during the study (no births, deaths or migration) • All individuals have an equal chance of being caught / animals are mobile and can be caught	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402, 405, 407; AQA spec 3.1.2.3.5-3.1.2.3.6.	
09.1	• The processes (eg regulation of atmospheric gases by photosynthesis and respiration) act in a dynamic equilibrium and cancel each other out • Because there is no overall change their actions go unnoticed / their importance is not appreciated until they are lost	2
09.2	<i>Any 3 from:</i> • Invertebrates, fungi and bacteria break down / decompose dead organic matter, releasing nutrients for plant growth • The organic matter and humus produced hold the soil together (reducing erosion) and regulate the water cycle, giving more even river flow and less flooding • Organic acids from decomposition aid the weathering of rock, producing more soil and releasing more nutrients • Microbes carry out the processes of the carbon, nitrogen and phosphorus cycles so wastes do not build up and nutrients are not depleted	3
09.3	• Trees reduce wind velocity so wind pollination would be ineffective / unreliable over long distances • Insects search over long distances for flowers so dispersed populations can be pollinated, and less pollen needs to be produced (energy saving); specialised flowers reduce pollen being wasted on other species	2
09.4	• Dispersal is not haphazard: animals live in the same habitat as the plant so are likely to carry seeds to locations where they can survive • Seeds can be carried longer distances than by wind and can be larger, containing more nutrients for the seedling • Seeds passing through the gut are deposited in faecal material which acts as a fertiliser, and passage through the intestine may stimulate germination	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 18-19; AQA spec 3.1.2.1.3.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Water: catchment management, conservation, metering, greywater, re-use of treated effluent, aquifer recharge, sustainable urban drainage - largely cyclical already; limits in arid regions • Minerals: exploiting low-grade ores with low-temperature processes (bioleaching), extending product lifespans, repair / re-use / repurpose / recycle, design for disassembly, reduced packaging; limits: dispersal, alloys, energy of collection • Waste management: repair, re-use, refurbish, recycle, composting and anaerobic digestion returning nutrients, minimal landfill, energy from residual waste; EU targets on packaging, batteries, WEEE • Pollution control: shift from post-production treatment to non-release - fuel desulfurisation not FGD, fuel cells / electric vehicles not combustion, industrial symbiosis (Kalundborg: waste heat, gypsum, fly ash, yeast used by neighbours); limits for legacy pollution • Energy: renewables, storage, energy density technologies, low-temperature manufacturing, CHP; limits of intermittency, transport fuels • Food: natural processes for nutrients (legumes, organic matter), pest control (IPM), soil biota, agri-environment schemes, IMTA aquaculture; tension with maximum yield for a growing population • Land uses supporting ecosystems: urban green infrastructure, gardens, permeable surfaces, habitat in redeveloped docks / lagoons • Judgement: high applicability in some sectors (water, waste), partial in others (minerals, energy for transport); needs policy support	9
10.2	• Locating industries so that the wastes of one (heat, gases, by-products) become the raw materials of another, mimicking natural cycles (eg Kalundborg eco-industrial park)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 375-379; AQA spec 3.6.4.1.	

Question	Marking guidance	Marks
11.1	Discuss the extent to which agriculture can be made sustainable while feeding a growing human population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Need: population possibly 12 billion; humans raise carrying capacity; trophic level of diet • Impacts to be reduced: habitat loss, drainage, nutrient enrichment, introduced species, pesticide and nutrient pollution, nitrate toxicity, greenhouse gases, hydrological changes, soil erosion • Social, economic and political influences: consumer choices, technology availability, grants, guaranteed markets, surpluses, quotas, set-aside, agri-environment schemes • Strategies: cultural and biological pest control, integrated control, organic nutrient supply, legumes, low tillage, water conservation, CWR conservation, renewable energy, methane reduction, habitat retention • Limits: little new land, imports reducing local availability, cost and acceptance of new techniques, past land degradation • Justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-337; AQA spec 3.5.1.3-3.5.1.5.	25

Question	Marking guidance	Marks
11.2	Discuss how human activities influence biodiversity, using examples from a range of different contexts. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Direct exploitation: food (fisheries, bushmeat), fashion, entertainment / pets, furniture and ornaments, traditional medicines, other products (whale oil); deliberate eradication of predators, competitors, vectors, pests • Changes in abiotic factors: water availability (drainage, abstraction, HEP), light (forest clearance), oxygen (organic pollution, thermal), nutrients (eutrophication), pH (acid mine drainage, acid rain), temperature (thermal pollution, climate change), turbidity, physical damage; inactivity (stopping plagioclimax management) • Changes in biotic factors: introduced competitors, predators, pathogens, hybridisation; loss of pollinators, seed dispersers, keystone species; food chain impacts (sea otters, puffins, jellyfish) • Habitat destruction: deforestation, farmland expansion, urbanisation, mining, reservoirs; fragmentation and edge effects • Contexts: temperate woodland, rainforest, coral reefs, oceanic islands, mangroves, Antarctica; agriculture, fishing, forestry • Positive influences: habitat creation, conservation management, reintroductions, legislation • Evaluation of which influences are most damaging and hardest to reverse; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26, 328-330, 341-344, 367-369; AQA spec 3.1.2.2, 3.5.1.3, 3.5.2.2, 3.5.3.3.	25